STACKS, DISCRIMINANTS: LEMMA-UNIVERSAL-FINITE-SYNTOMIC-SMOOTH

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2383-2388 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2390-2415, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_discriminant.json (pb=1426, stmt 2383-2388, proof 2390-2415).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 051F: sources/stacks-discriminant/pinned-original-source.tex, source lines 2415–2415. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FKZ: sources/stacks-discriminant/pinned-original-source.tex, source lines 2326–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FL0: sources/stacks-discriminant/pinned-original-source.tex, source lines 2363–2368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02HX: sources/stacks-discriminant/pinned-original-source.tex, source lines 2415–2415. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07M8: sources/stacks-discriminant/pinned-original-source.tex, source lines 2415–2415. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *With notation as in Example 0FKZ and U_d as in Lemma 0FL0 then U_d is smooth over* $\mathrm{Spec}(\mathbf{Z})$.

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let us use More on Morphisms, Lemma 02HX to show that $U_d \to \mathrm{Spec}(\mathbf{Z})$ is smooth. Namely, suppose that $\mathrm{Spec}(A) \to U_d$ is a morphism and $A' \to A$ is a small extension. Then $B = A \otimes_{A_d} B_d$ is a finite free A-algebra which is syntomic over A (by construction of U_d). By Smoothing Ring Maps, Proposition 07M8 there exists a syntomic ring map $A' \to B'$ such that $B \cong B' \otimes_{A'} A$. Set $e'_1 = 1 \in B'$. For $1 < i \leq d$ choose lifts $e'_i \in B'$ of the elements $1 \otimes e_i \in A \otimes_{A_d} B_d = B$. Then $e'_1, \ldots, e'_d$ is a basis for B' over A' (for example see Algebra, Lemma 051F). Thus we can write $e'_i e'_j = \sum \alpha_{ij}^l e'_l$ for unique elements $\alpha_{ij}^l \in A'$ which satisfy the relations $\sum_l \alpha_{ij}^l \alpha_{lk}^m = \sum_l \alpha_{il}^m \alpha_{jk}^l$ and $\alpha_{ij}^k = \alpha_{ji}^k$ and $\alpha_{i1}^j = \delta_{ij}$ in A'. This determines a morphism $\mathrm{Spec}(A') \to X_d$ by sending $a_{ij}^l \in A_d$ to $\alpha_{ij}^l \in A'$. This morphism agrees with the given morphism $\mathrm{Spec}(A) \to U_d$. Since $\mathrm{Spec}(A')$ and $\mathrm{Spec}(A)$ have the same underlying topological space, we see that we obtain the desired lift $\mathrm{Spec}(A') \to U_d$ and we conclude that U_d is smooth over $\mathbf{Z}$. $\square$